DAY 04 • STUDENT ACTIVITIES

Security Solutions, Detection Sources and Log Management

Where alerts come from, how security tools classify severity, and how logs are collected and read

Qualification	Cyber Threat Monitoring Level I
Unit of Competency	CS-ICT251101 Monitor and Report Cyber Threats
Coverage	Range of Variables: Detection alert sources and Security solutions; Required Knowledge 1.5 Sources of detection alerts, 1.6 Log and detection management, 1.8 Security solution management application, 1.9 Severity classifications
Duration	8 hours (1 training day)

Trainee Name	Batch / Section	Date

Activities Overview

No.	Activity	Type	Time	Assets Needed
4.1	Which Tool Would Catch It?	Pair	30 min	None
4.2	Identify the Alert Source	Individual	45 min	multi_source_alerts.csv (lab: ~/day4)
4.3	Log Hunt: Brute Force to Compromise	Individual (hands-on)	75 min	firewall.log, windows_security_events.csv (lab: ~/day4)
4.4	Explore a Security Console	Individual (hands-on)	45 min	CTM Virtual Lab: your WKS-LAB workstation (PowerShell); Wazuh dashboard if available

CAUTION — Lab Rules

- Work only on the lab machines, virtual machines and files provided by your trainer.
- Never scan, probe or test any system outside the authorized training network.
- Treat all sample logs and data as confidential, as if they belonged to a real client.

Activity 4.1 — Which Tool Would Catch It?

Type	Pair
Time Allotment	30 min
Objective	Match threat scenarios with the security solution most likely to detect them.
Assets / Materials	None (worksheet only)

Instructions

1. For each scenario, choose the best detecting solution and a second layer.
2. Be ready to defend your choice.

Work Space

Scenario	Primary Solution	Second Layer
An attacker sends SQL injection to the online banking login page		
An employee emails a spreadsheet of customer TINs to a personal account		
A laptop runs a macro that launches PowerShell		
A CCTV camera starts scanning the internal network		
Someone in the internet tries RDP on the public IP		
20 failed VPN logins, then a success, then a file-server login at 2 AM		
A water utility PLC receives a command from an unknown workstation		
A user downloads a known malicious file		

Deliverable

Completed matching table (worksheet in your CTM Virtual Lab notebook).

Performance Criteria (Trainer Use)

Criteria	Yes	No
Primary solutions correctly identified for at least 6 of 8	☐	☐
Can explain defense in depth	☐	☐

Activity 4.2 — Identify the Alert Source

Type	Individual
Time Allotment	45 min
Objective	Determine the detection source and key fields of alerts from a SIEM export.
Assets / Materials	assets/multi_source_alerts.csv (CTM Virtual Lab: ~/day4/)

Instructions

1. Open assets/multi_source_alerts.csv in a spreadsheet program, or in the CTM Virtual Lab (~/day4/) with Import-Csv in PowerShell or cut in Ubuntu.
2. For each alert, identify the detection source (SIEM, AV, Firewall, WAF, DLP, EDR, NDR).
3. Record the affected asset and the normalized severity (Critical/High/Medium/Low/Info).
4. Sort the alerts from most to least urgent.

Work Space

Alert ID	Source	Affected Asset	Normalized Severity	Urgency Rank

Deliverable

Completed table with urgency ranking (worksheet in your CTM Virtual Lab notebook).

Performance Criteria (Trainer Use)

Criteria	Yes	No
Sources correctly identified	☐	☐
Vendor severities correctly normalized	☐	☐

Criteria	Yes	No
Ranking puts ransomware and active malware at the top	☐	☐

Activity 4.3 — Log Hunt: Brute Force to Compromise

Type	Individual (hands-on)
Time Allotment	75 min
Objective	Find a brute-force pattern and its outcome across firewall and Windows logs.
Assets / Materials	assets/firewall.log; assets/windows_security_events.csv (CTM Virtual Lab: ~/day4/)

Instructions

1. Open assets/firewall.log. Find the source IP with the most connections to TCP 3389.
2. Open assets/windows_security_events.csv. Filter Event ID 4625 (failed logon) and 4624 (successful logon).
3. Determine whether the attacker eventually logged in, with which account, and at what time.
4. List what happened after the successful logon (look at Event IDs 4688, 4720, 4732).
5. In the CTM Virtual Lab (Ubuntu tab) you may use: `grep ':3389 ' ~/day4/firewall.log | awk '{print $6}' | cut -d= -f2 | cut -d: -f1 | sort | uniq -c`.

Work Space

Attacker source IP	
Number of failed attempts	
Targeted account(s)	
Successful logon time and account	
Actions after logon	
Kill Chain phase(s) reached	

Deliverable

Completed findings table (worksheet in your CTM Virtual Lab notebook).

Performance Criteria (Trainer Use)

Criteria	Yes	No
Identified the attacking IP	☐	☐
Correctly counted failed attempts	☐	☐
Found the successful logon and follow-on actions (new account, admin group)	☐	☐
Concluded that the host is compromised	☐	☐

Activity 4.4 — Explore a Security Console

Type	Individual (hands-on)
Time Allotment	45 min
Objective	Locate alerts, status and logs in real security solution consoles.
Assets / Materials	CTM Virtual Lab: your WKS-LAB workstation (Get-MpThreat, Get-WinEvent, Get-Service); Windows Security and Event Viewer on a real Windows PC; Wazuh dashboard if available

Instructions

1. Protection history: in the CTM Virtual Lab run Get-MpThreat and Get-MpThreatDetection (on a real Windows PC: Windows Security > Virus & threat protection > Protection history). Record the latest entries.
2. Defender log: in the lab run Get-WinEvent -FilterHashtable @{LogName='Microsoft-Windows-Windows Defender/Operational'; Id=1116,1117} (Event Viewer > Applications and Services Logs > Microsoft > Windows > Windows Defender > Operational). Find Event IDs 1116/1117 or 1000/1001.
3. Security log: in the lab run Get-WinEvent -FilterHashtable @{LogName='Security'; Id=4624} (Event Viewer > Windows Logs > Security, filter Event ID 4624) and record your own logon.
4. Agent status: in the lab run Get-Service Wazuh*. If the trainer provides the Wazuh dashboard, also find the alerts list and an agent's status.

Work Space

Location	What I Found

Deliverable

Console exploration notes.

Performance Criteria (Trainer Use)

Criteria	Yes	No
Found the Protection history and event log data (lab PowerShell or Windows consoles)	☐	☐
Applied an Event ID filter	☐	☐

Self-Check

A. Multiple Choice

Encircle the letter of the best answer.

1. Which solution protects websites from SQL injection?
    a) a. DLP
    b) b. WAF
    c) c. NDR
    d) d. SOAR
2. Which solution automates response playbooks?
    a) a. SIEM
    b) b. SOAR
    c) c. IDS
    d) d. AV
3. An IDS differs from an IPS because an IDS:
    a) a. Blocks traffic
    b) b. Only detects and alerts
    c) c. Encrypts traffic
    d) d. Scans files
4. Converting logs into common fields is called:
    a) a. Correlation
    b) b. Normalization
    c) c. Retention
    d) d. Aggregation
5. A DLP alert most likely concerns:
    a) a. Port scanning
    b) b. Sensitive data leaving the organization
    c) c. Malware signatures
    d) d. CPU usage

B. True or False

Write TRUE if the statement is correct, FALSE if it is not.

No.	Statement	Answer
1	A tool's severity rating is always final.	
2	XDR combines data from endpoints, network, email and cloud.	
3	The Philippine time zone is UTC+8.	
4	MDR is security monitoring provided by an external provider.	

C. Short Answer

1. List the six steps of the log pipeline.

2. Give two reasons why log timestamps and time zones are important.

Reflection

What was the most important thing you learned today, and how will you apply it in a Security Operations Center?

Trainer Sign-off

Overall Result	Trainer Remarks	Trainer Signature / Date
☐ Competent		
☐ Not Yet Competent		